POLITECNICO
MILANO 1863

Classification of Pumpkin Seeds by using Machine Learning

Data Analytics For Smart Agriculture

Table of Contents

Pumpkin Lovers!

Our team consists of 4 people:

Negar Najafi

Mostafa Hashemiyan

Pouria Saadaty

Nadia Sadeghi

Master students of Telecommunication
engineering from Iran

Data background and motivation:

Data background and motivation:

All about Pumpkin seeds:

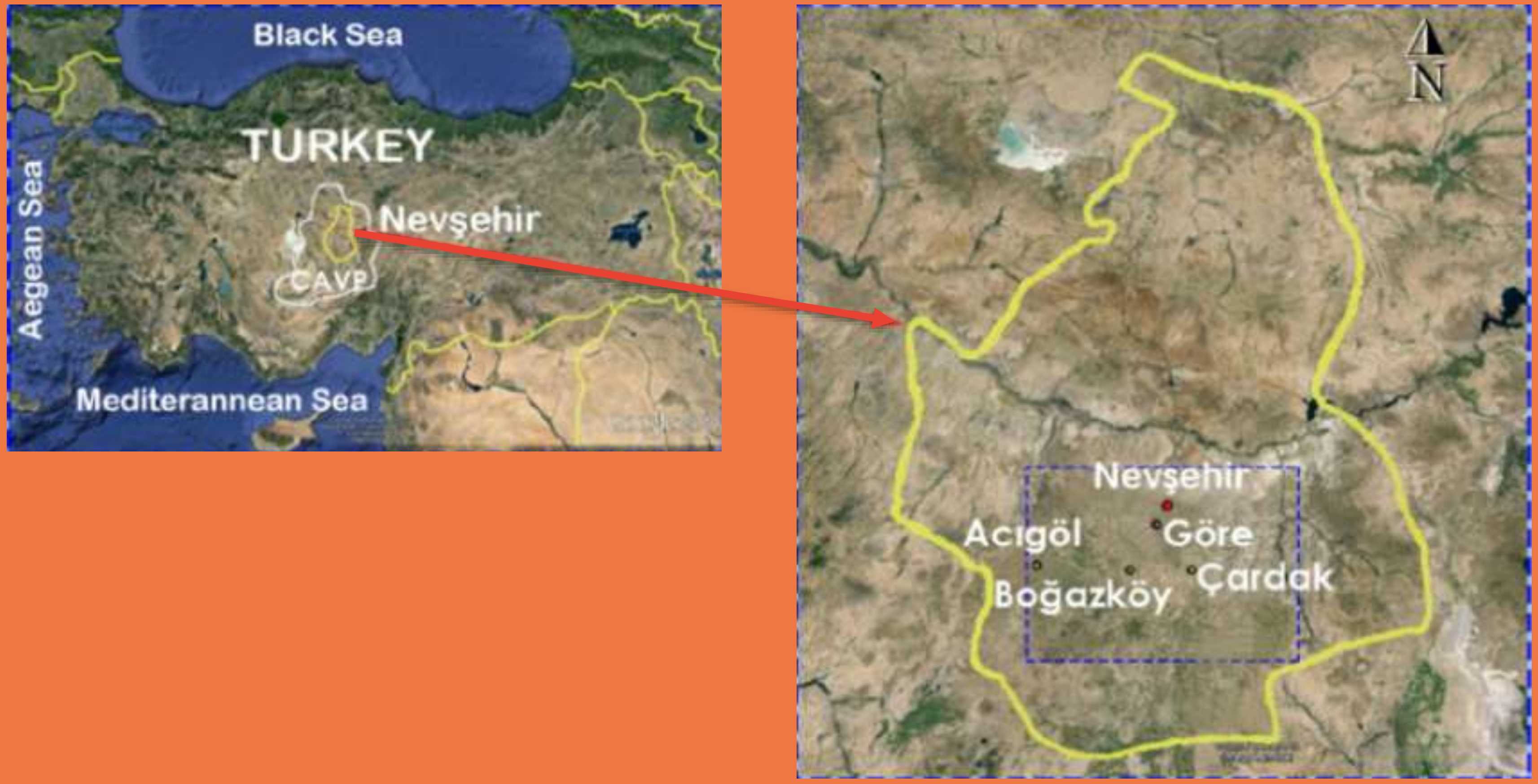
Two main pumpkin seed varieties in Turkey:

Ürgüp Sivrisi (class A)

Çerçevelik (class B)

•Both are widely grown and used in confectionery

•Traditional visual inspection is subjective and time-consuming



The Path

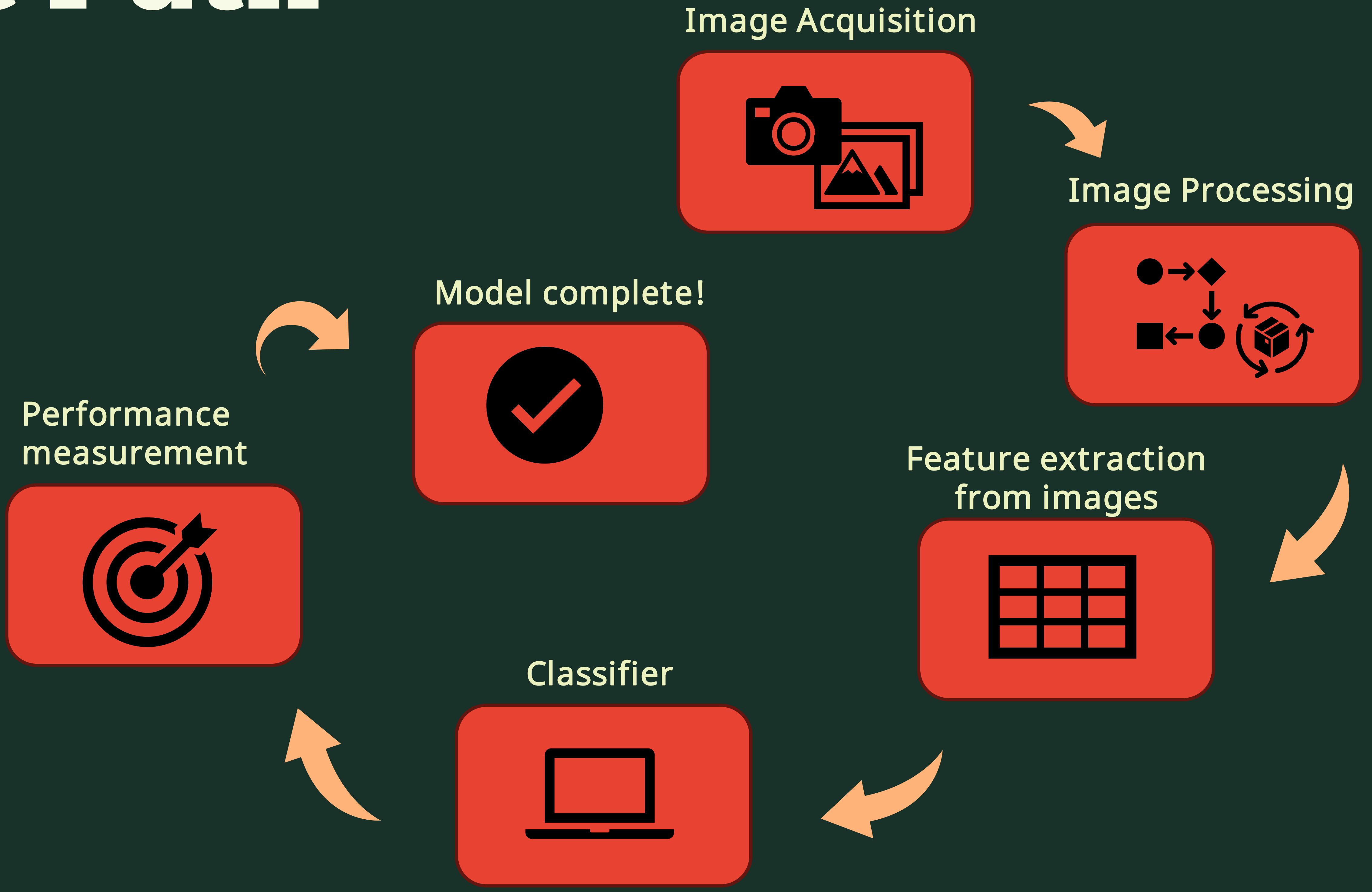
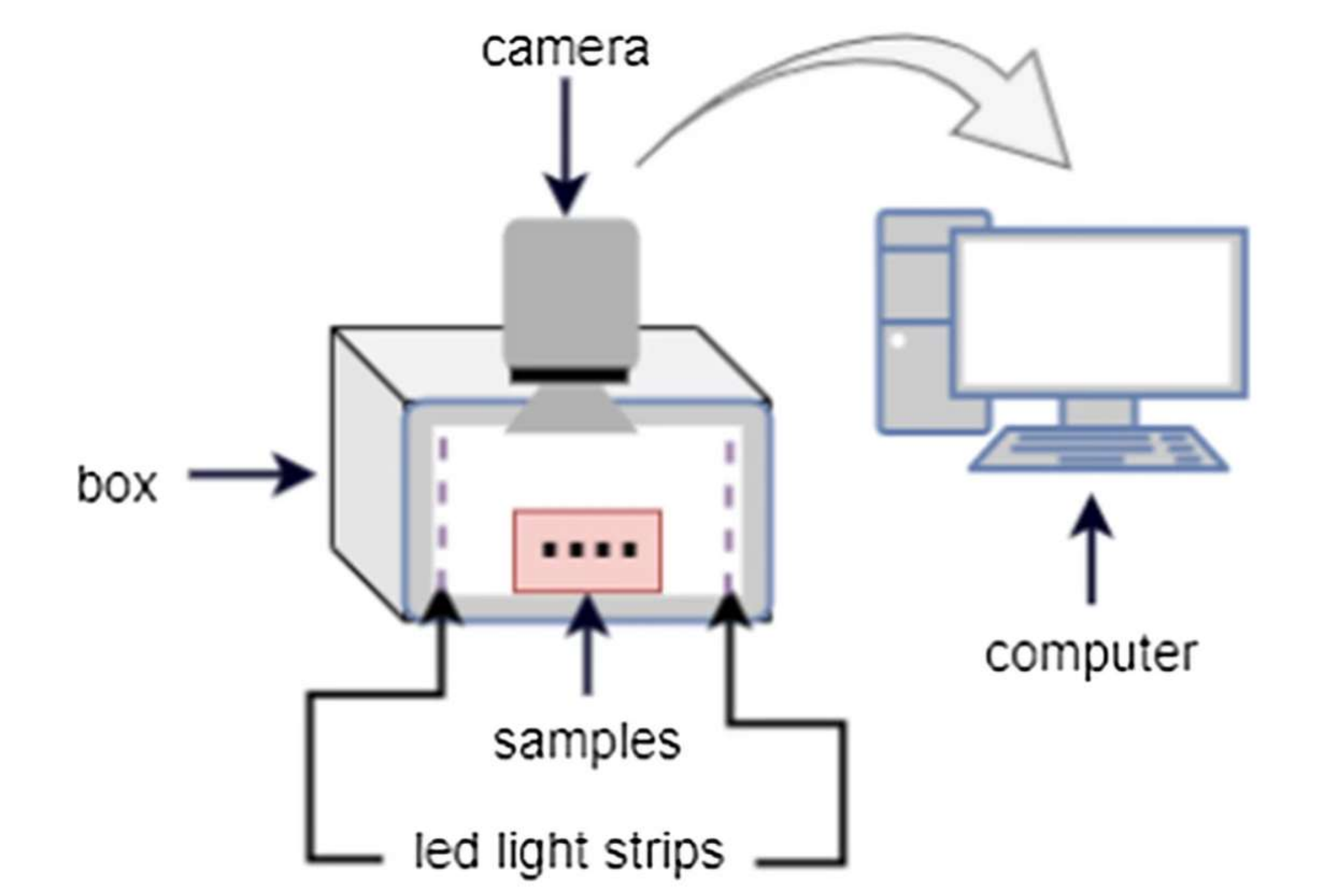


Image Acquisition
Image Processing
Feature extraction
from images
Classifier
Performance
measurement
Model complete!

Image acquisition:




(a) Ürgüp Sivrisi

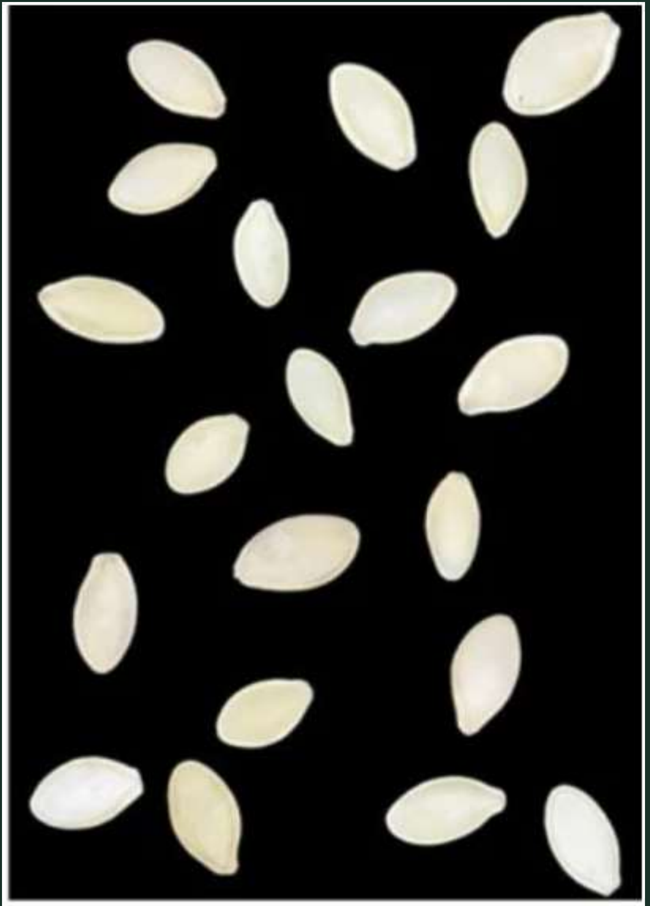
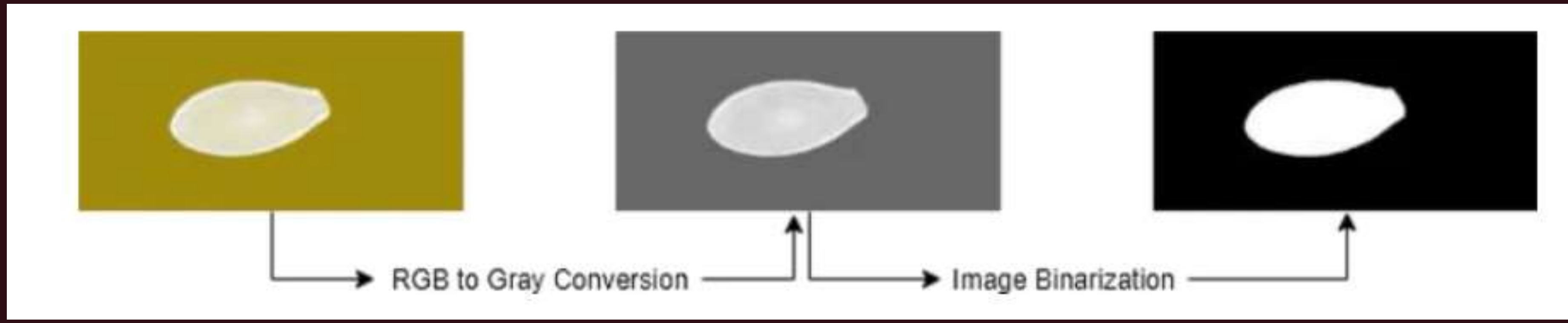
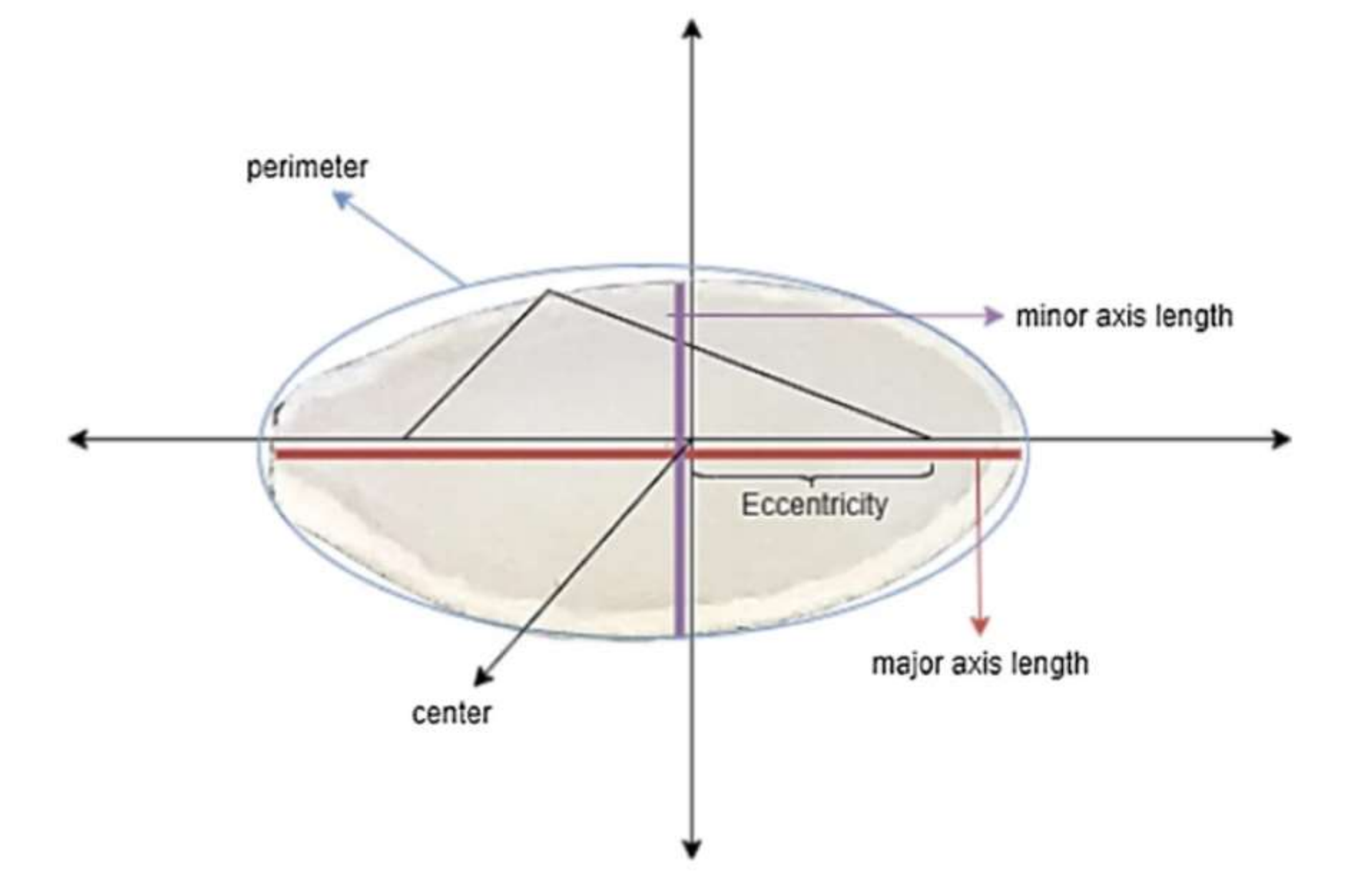

(b) Çerçevelik

Image processing



Steps of image processing



The sampling of pumpkin seeds on a 2-D plane



DATA PROCESS




WHAT DO WE HAVE?

1

DATA EXPLORATION

analyzing a dataset related to pumpkin seeds, which contains **2,500 samples**

2

WHAT IS THE GOAL OF THIS PROJECT

The goal is to classify different types of pumpkin seeds based on their shape and size features

3

FEATURE SELECTION

Base on the data correlation and overlap in features ,remove them and reduce the feature to increase performance of the model

Dataset Overview

Feature	**Description**
Area	Total surface area of the seed
Perimeter	The boundary length of the seed
Major/Minor Axis Length	Longest & shortest axis of the seed
Convex Area	Area of the convex hull enclosing the seed
Eccentricity	Measures how elongated the seed is (0 to 1)
Solidity	Ratio of seed's area to convex area
Extent	Bounding box area vs. actual area
Roundness	How circular the seed shape is
Aspect Ratio	Major axis length / Minor axis length
Compactness	Shape compactness measure
Class	Categorical variable indicating seed type

DATA EXPLORATION

	Area	Perimeter	Major_Axis_Length	Minor_Axis_Length	Convex_Area	Equiv_Diameter	Eccentricity	Solidity	Extent	Roundness	Aspect_Ration	Compactness	
0	56276	888.242	326.1485	220.2388	56831	267.6805	0.7376	0.9902	0.7453	0.8963	1.4809	0.8207	Çerçe
1	76631	1068.146	417.1932	234.2289	77280	312.3614	0.8275	0.9916	0.7151	0.8440	1.7811	0.7487	Çerçe
2	71623	1082.987	435.8328	211.0457	72663	301.9822	0.8749	0.9857	0.7400	0.7674	2.0651	0.6929	Çerçe
3	66458	992.051	381.5638	222.5322	67118	290.8899	0.8123	0.9902	0.7396	0.8486	1.7146	0.7624	Çerçe
4	66107	998.146	383.8883	220.4545	67117	290.1207	0.8187	0.9850	0.6752	0.8338	1.7413	0.7557	Çerçe

NUMBER OF ROWS
2500

NUMBER OF COLUMNS
13

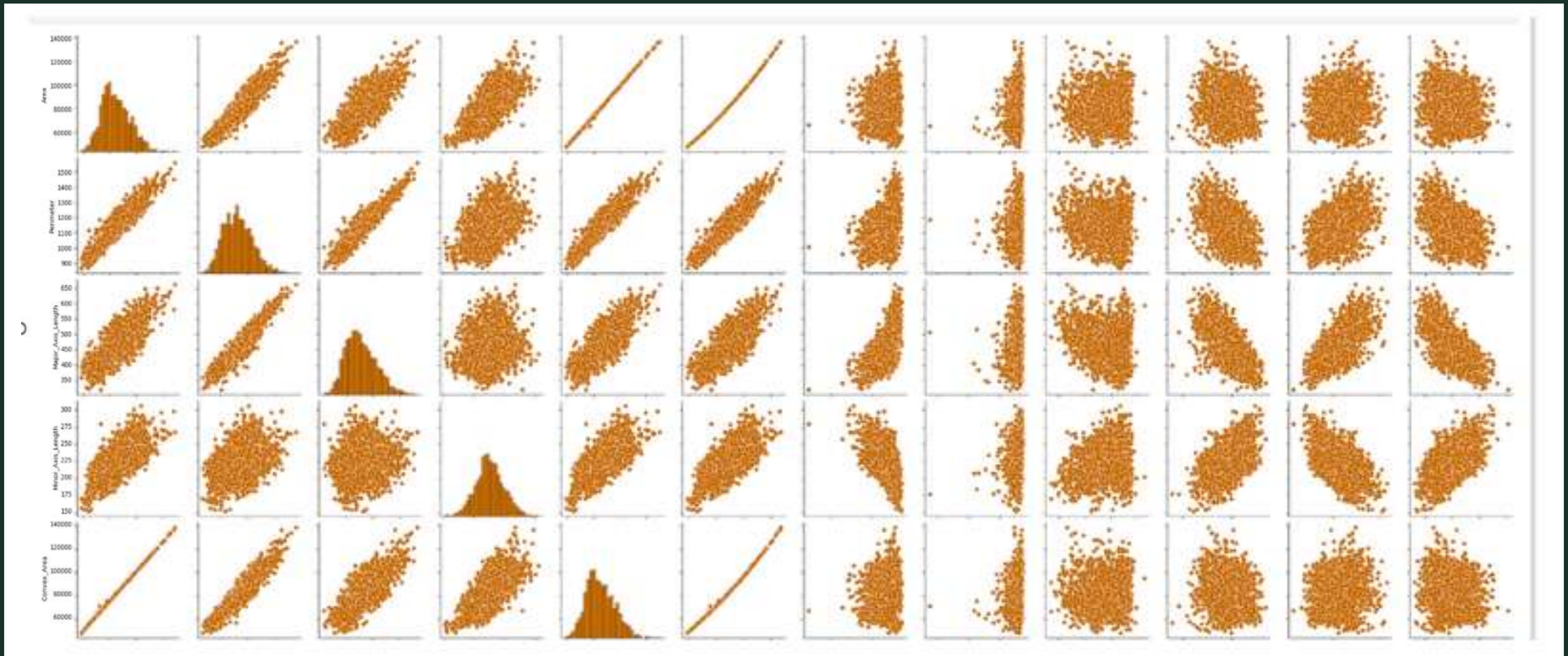
TARGET VALUE

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2500 entries, 0 to 2499
Data columns (total 13 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   Area               2500 non-null   int64  
 1   Perimeter          2500 non-null   float64
 2   Major_Axis_Length  2500 non-null   float64
 3   Minor_Axis_Length  2500 non-null   float64
 4   Convex_Area        2500 non-null   int64  
 5   Equiv_Diameter     2500 non-null   float64
 6   Eccentricity       2500 non-null   float64
 7   Solidity           2500 non-null   float64
 8   Extent             2500 non-null   float64
 9   Roundness          2500 non-null   float64
 10  Aspect_Ration      2500 non-null   float64
 11  Compactness        2500 non-null   float64
 12  Class              2500 non-null   object 
dtypes: float64(10), int64(2), object(1)
memory usage: 254.0+ KB
```

📌The dataset has a categorical label (Class) that indicates the type of pumpkin seed.

📌Our goal is to predict this class based on the numerical features.

PAIR PLOT

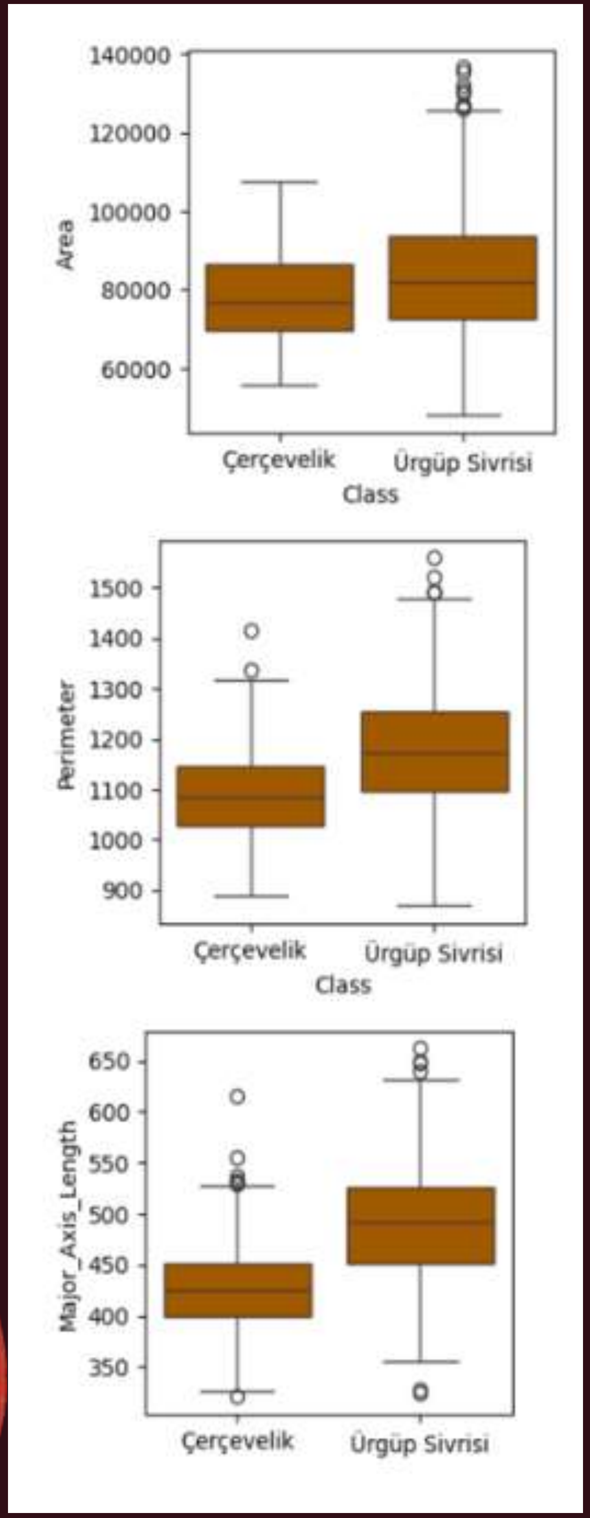


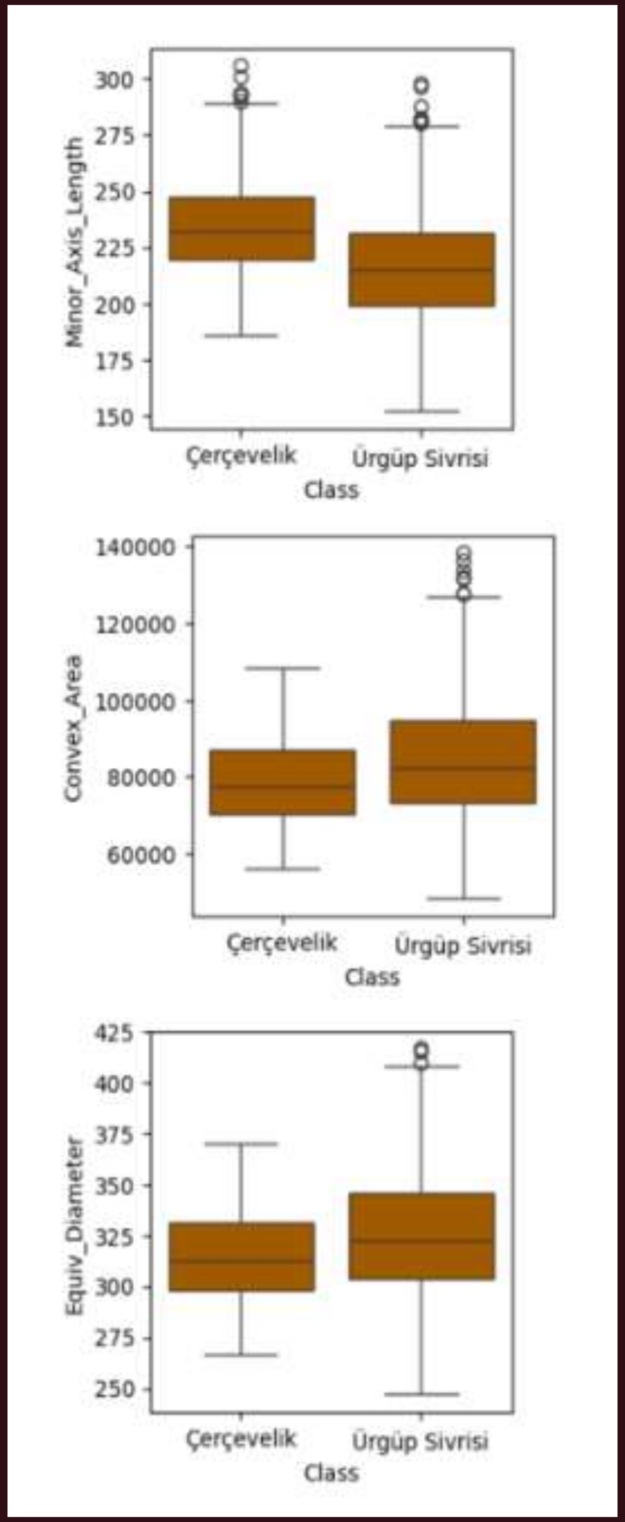
Linear Relationships

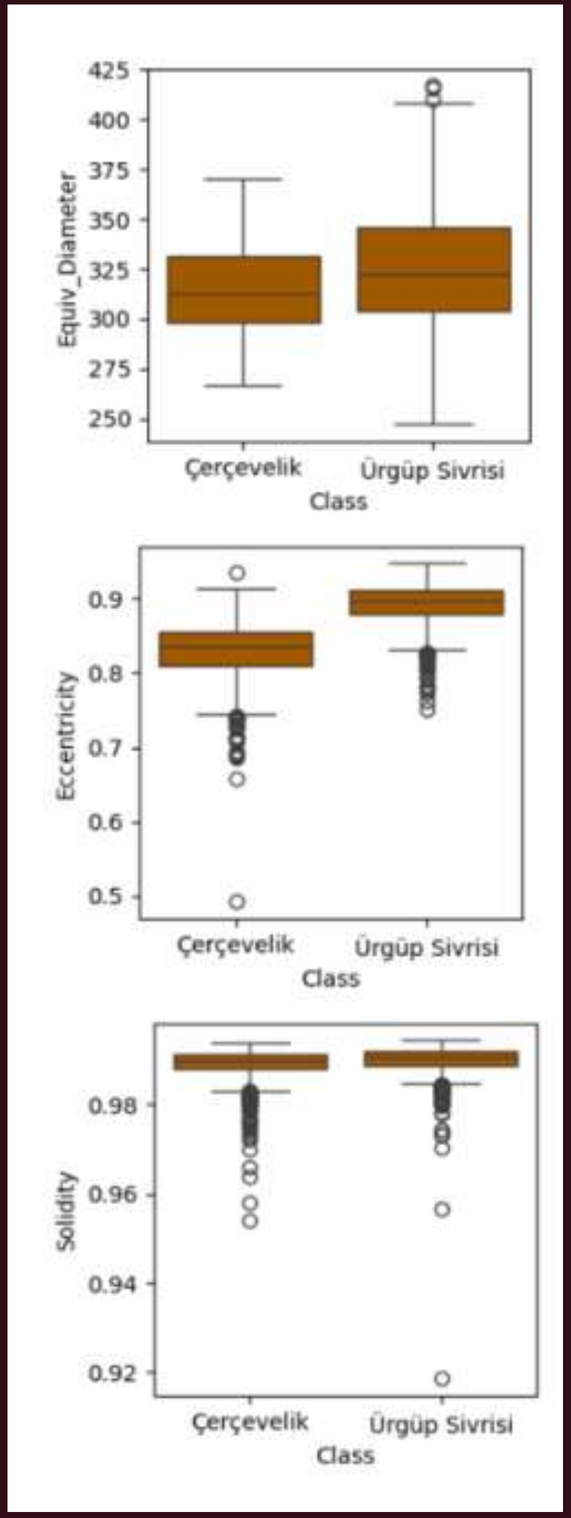
Dense Clusters

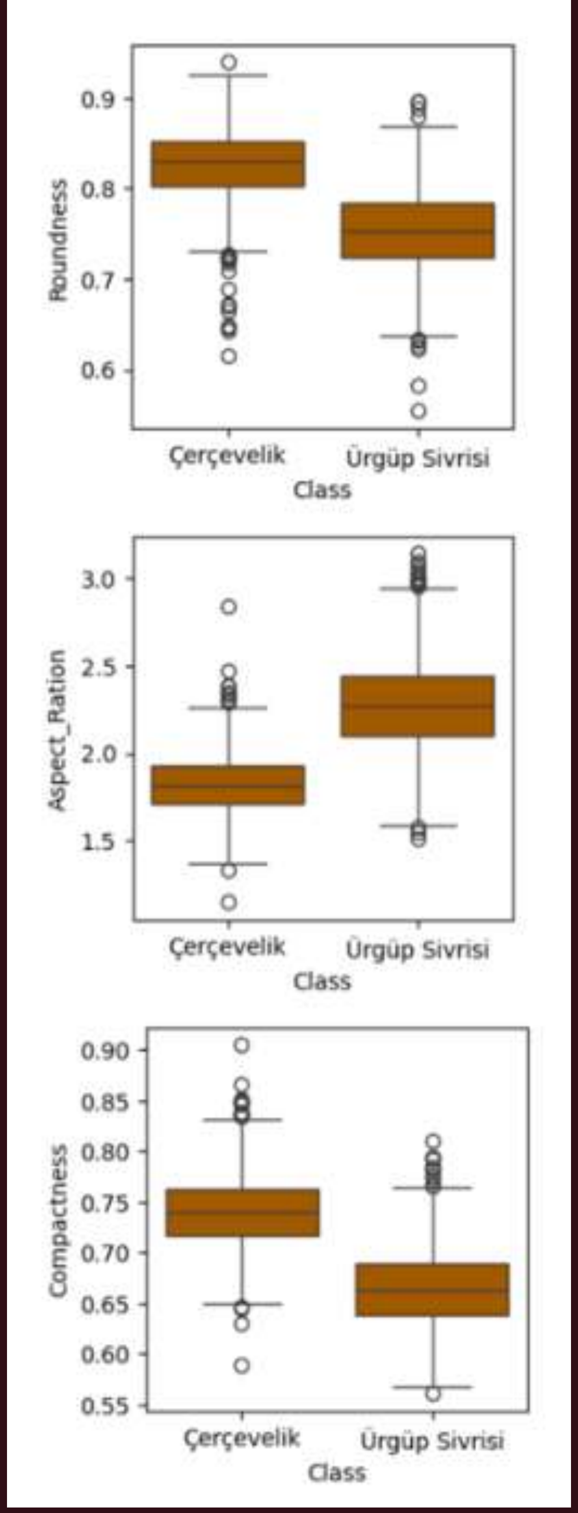
Variability

BOX PLOT

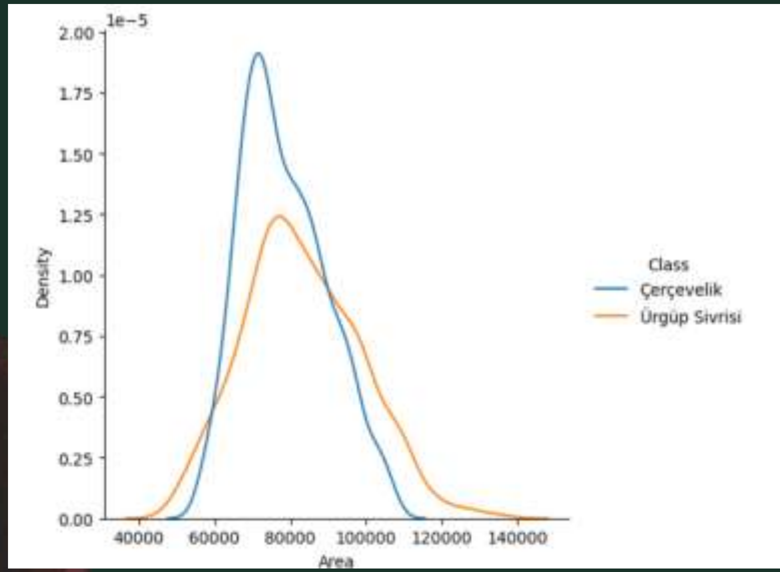

Area
Perimeter
Major_Axis_Length
Çerçevelik
Ürgüp Sivrisi
Class


Minor_Axis_Length
Convex_Area
Equiv_Diameter
Çerçevelik
Ürgüp Sivrisi
Class

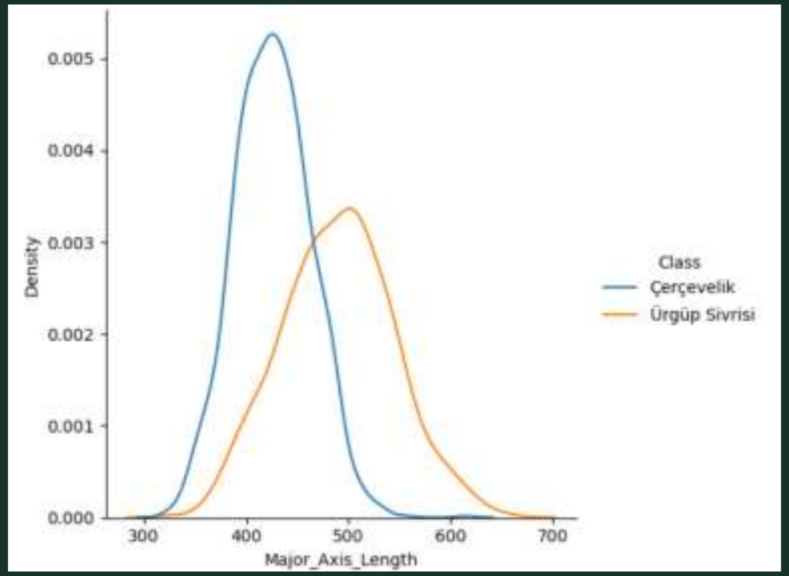

Equiv_Diameter
Eccentricity
Solidity
Çerçevelik
Ürgüp Sivrisi
Class


Roundness
Aspect_Ration
Compactness
Çerçevelik
Ürgüp Sivrisi
Class

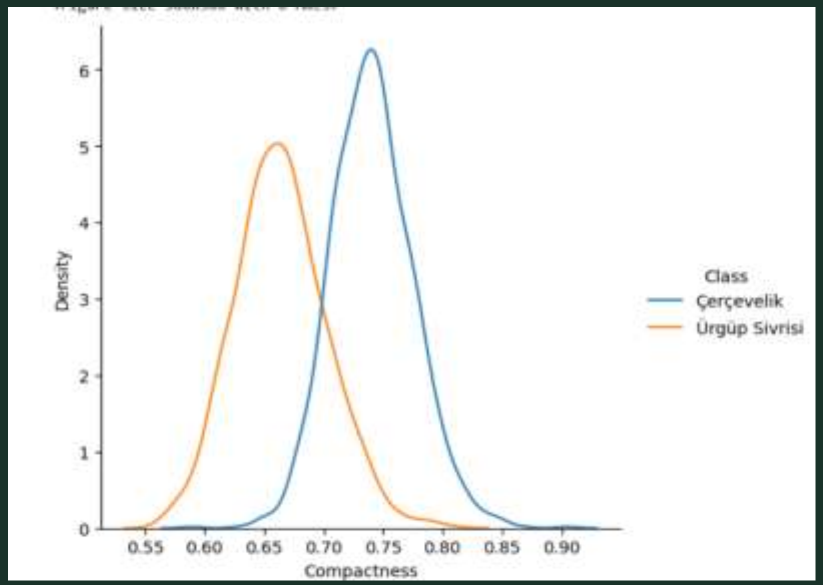
SIMILARITY



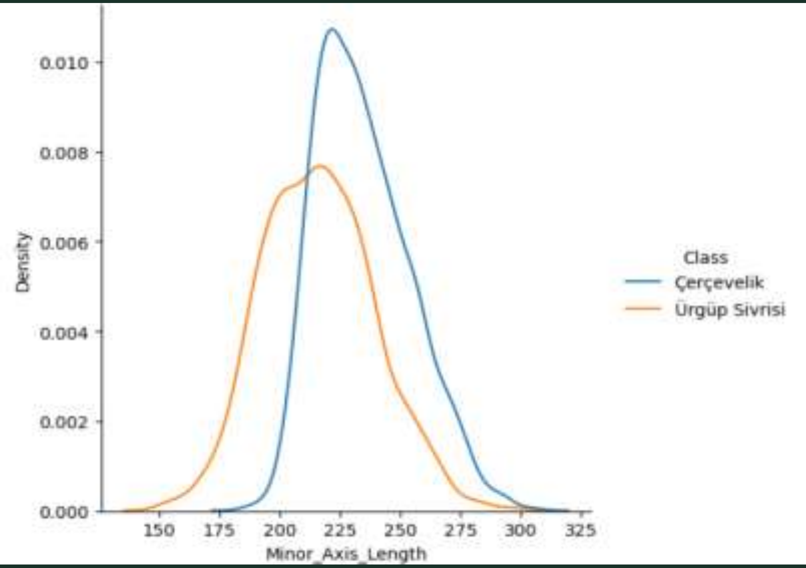
1e−5
Density
Area
Class
Çerçevelik
Ürgüp Sivrisi



Density
Major_Axis_Length
Class
Çerçevelik
Ürgüp Sivrisi



Density
Compactness
Class
Çerçevelik
Ürgüp Sivrisi

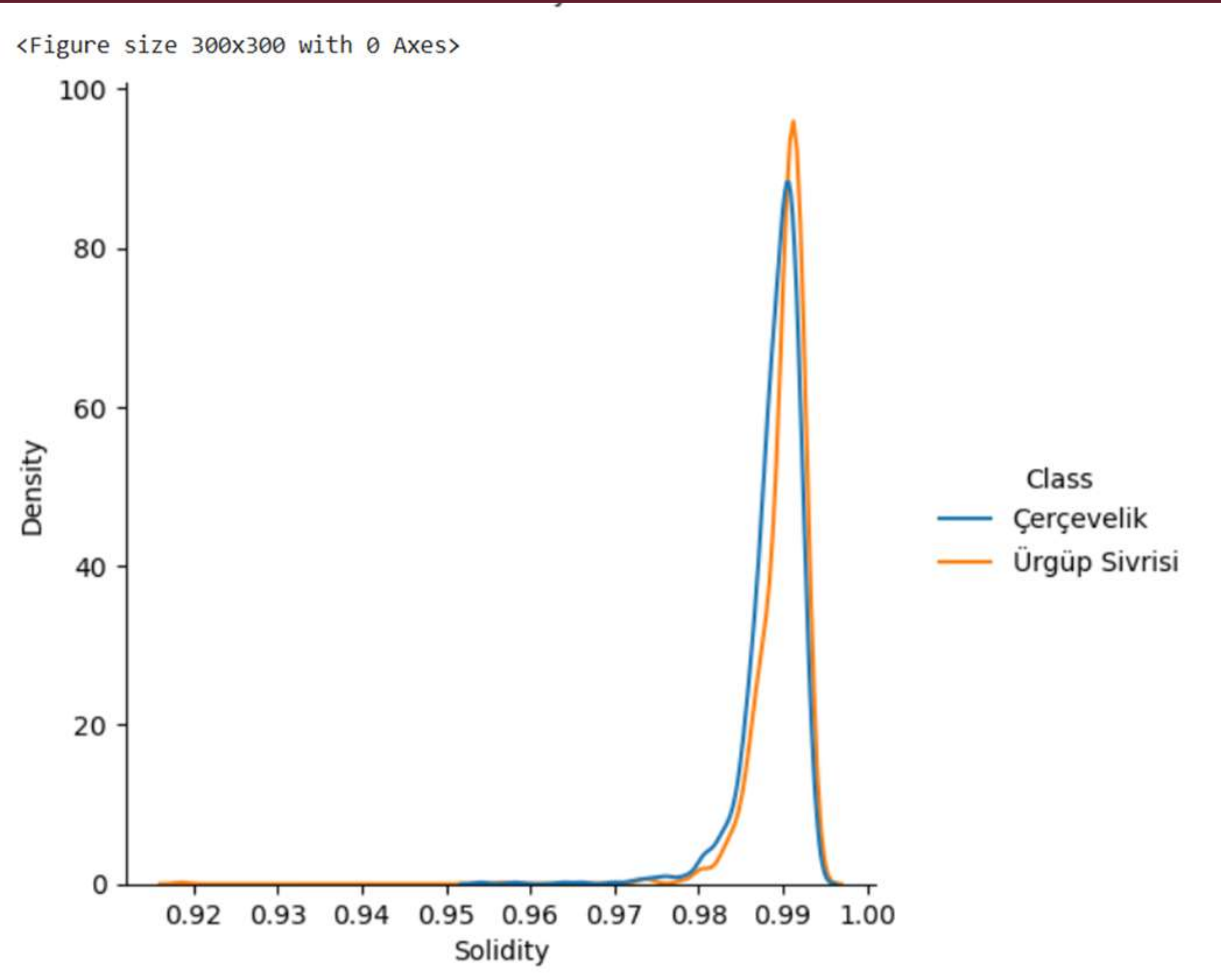


Density
Minor_Axis_Length
Class
Çerçevelik
Ürgüp Sivrisi

DATA OVERLAP

📌Some features may be irrelevant, redundant, or highly similar, leading to overfitting and slower model performance

📌By removing unnecessary features, we improve accuracy, reduce complexity, and make the model more efficient



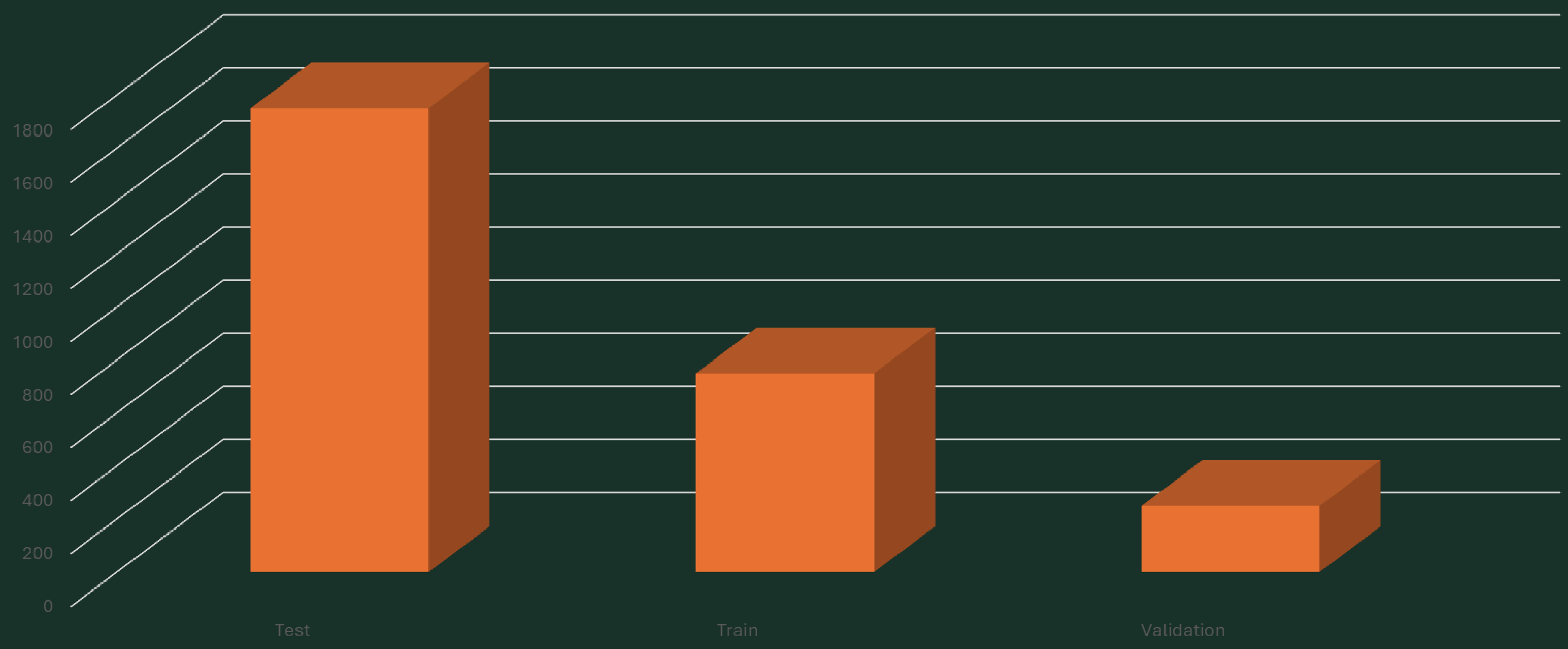
EXAMPLE DATA AFTER CLEANING

[13]:

	Perimeter	Major_Axis_Length	Minor_Axis_Length	Eccentricity	Extent	Roundness	Aspect_Ration	Compactness	Class	C_Class
0	888.242	326.1485	220.2388	0.7376	0.7453	0.8963	1.4809	0.8207	0	Çerçevelik
1	1068.146	417.1932	234.2289	0.8275	0.7151	0.8440	1.7811	0.7487	0	Çerçevelik
2	1082.987	435.8328	211.0457	0.8749	0.7400	0.7674	2.0651	0.6929	0	Çerçevelik
3	992.051	381.5638	222.5322	0.8123	0.7396	0.8486	1.7146	0.7624	0	Çerçevelik
4	998.146	383.8883	220.4545	0.8187	0.6752	0.8338	1.7413	0.7557	0	Çerçevelik

REDUCE TO 8 FETURE

DATA SPLITING



TRAIN **TEST**



MODELING




MODELING

To analyze the impact of feature selection on classification, we applied multiple machine learning models and compared their performance.

Each model has different strengths, from simple linear methods to complex neural networks.

4 MODEL USES

Models

Logistic regression

K-nearest neighbor

Random forest

Neural network

Gradient Boosting (GBN)

Hyperparameter search

K - fold

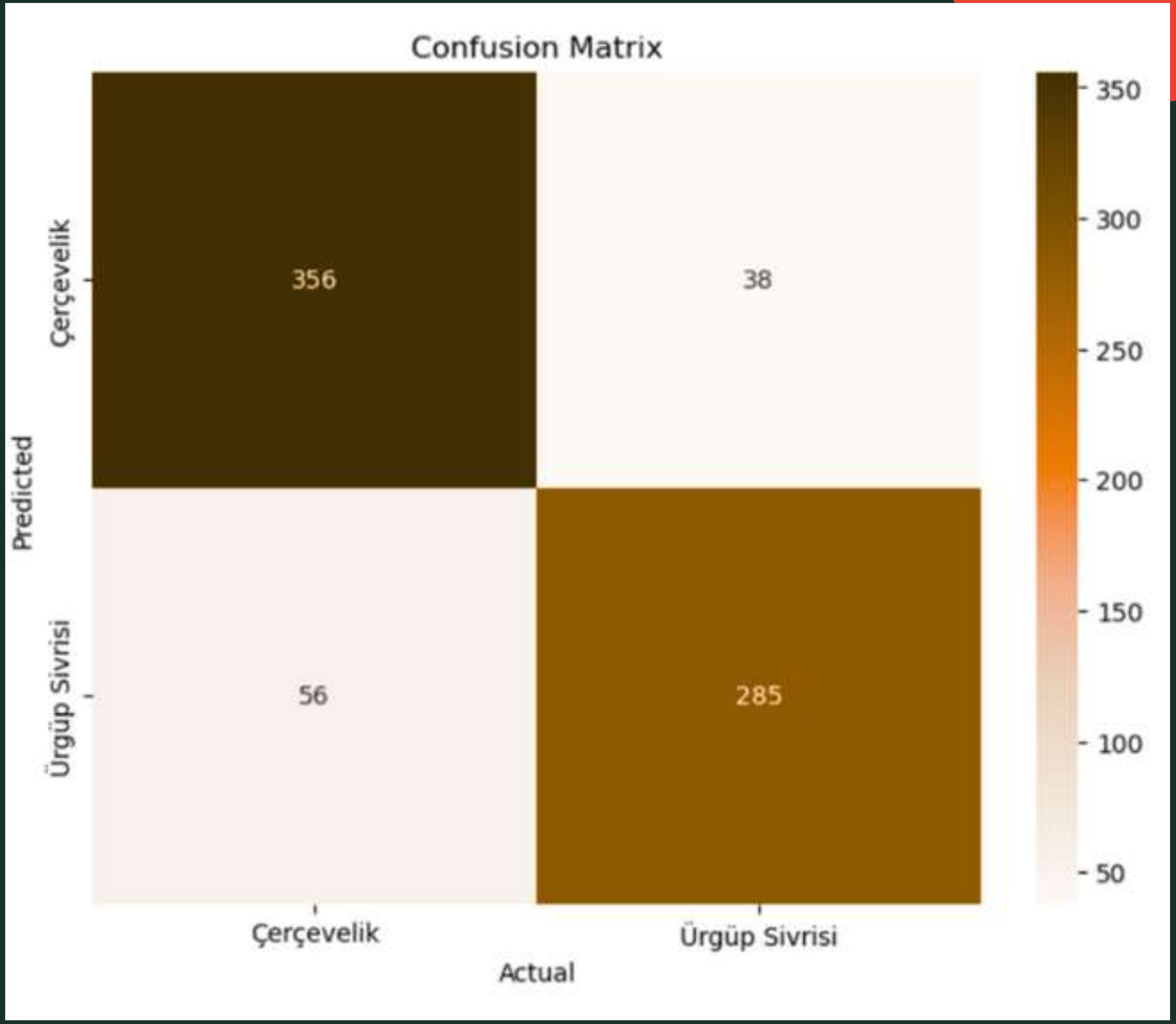
X

[Features] → Y [class]

Performances

Average Accuracy & F1-score

LOGESTIC REGRESSION



Accuracy
87.21 %

F1 score
87.18%

CONFIUSON MATRXI

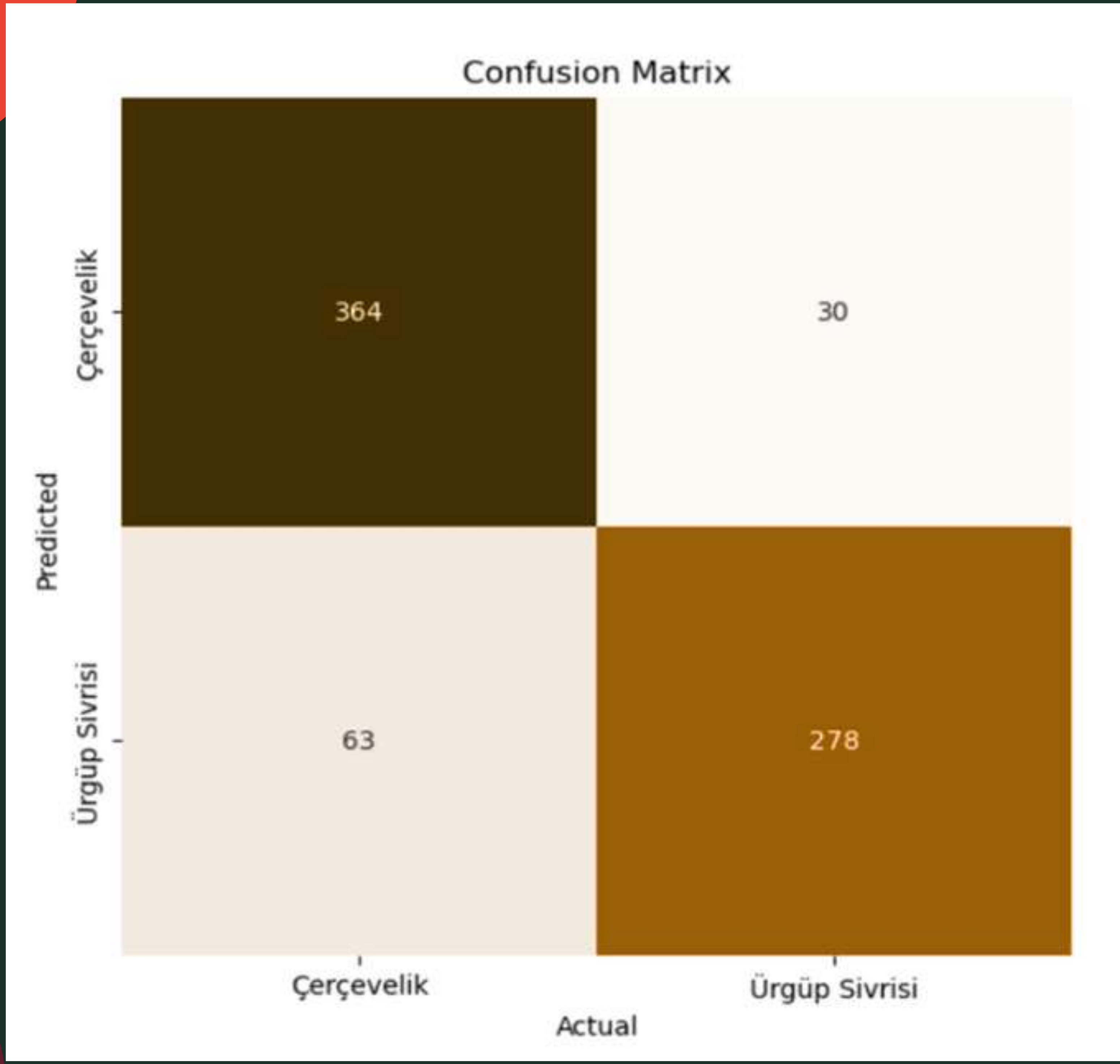
K-Nearest Neighbors (KNN)

Accuracy

87.34 %

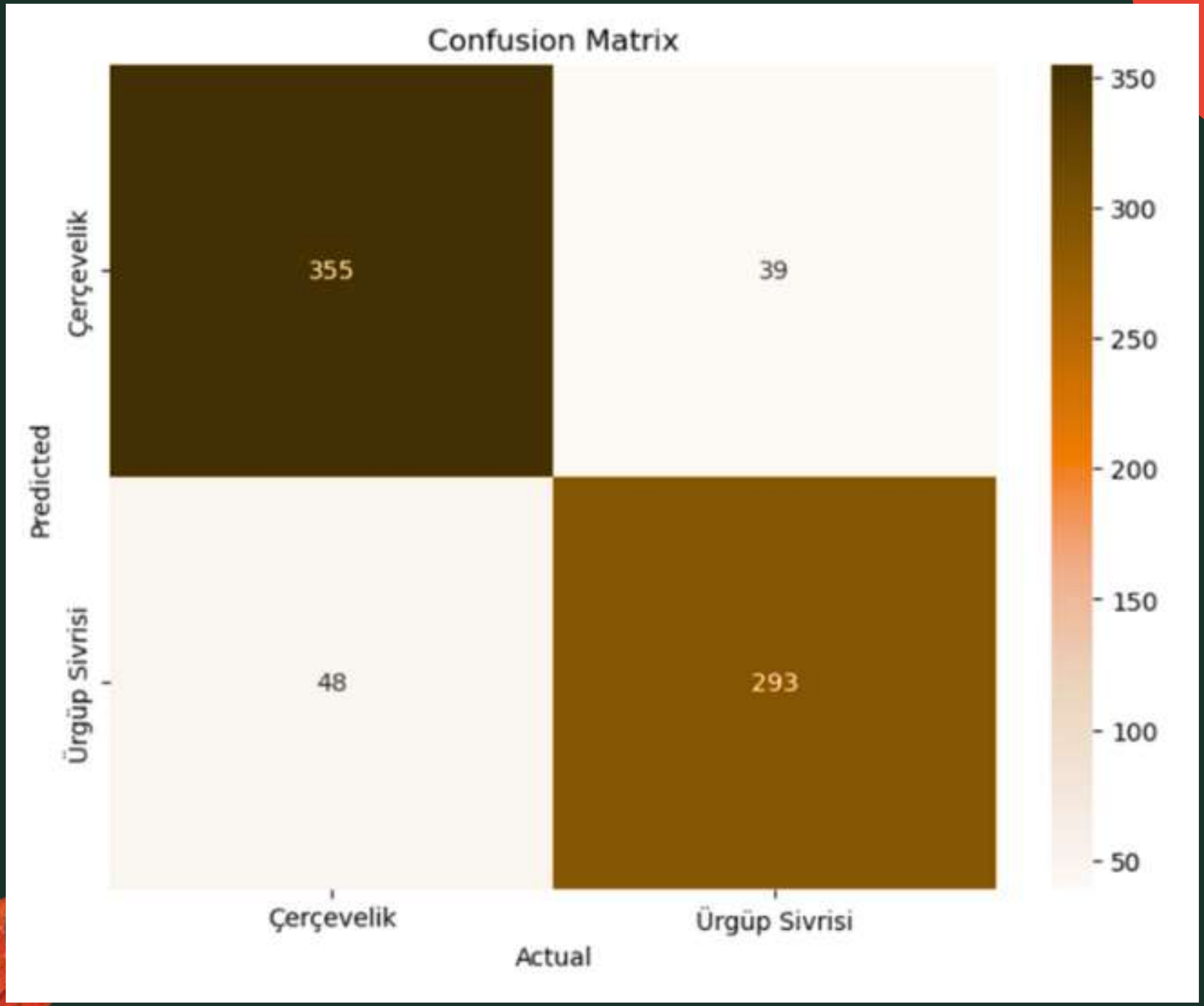
F1 score

87.28%



CONFIUSON MATRXI

RANDOM FOREST



Accuracy

88.16 %

F1 score

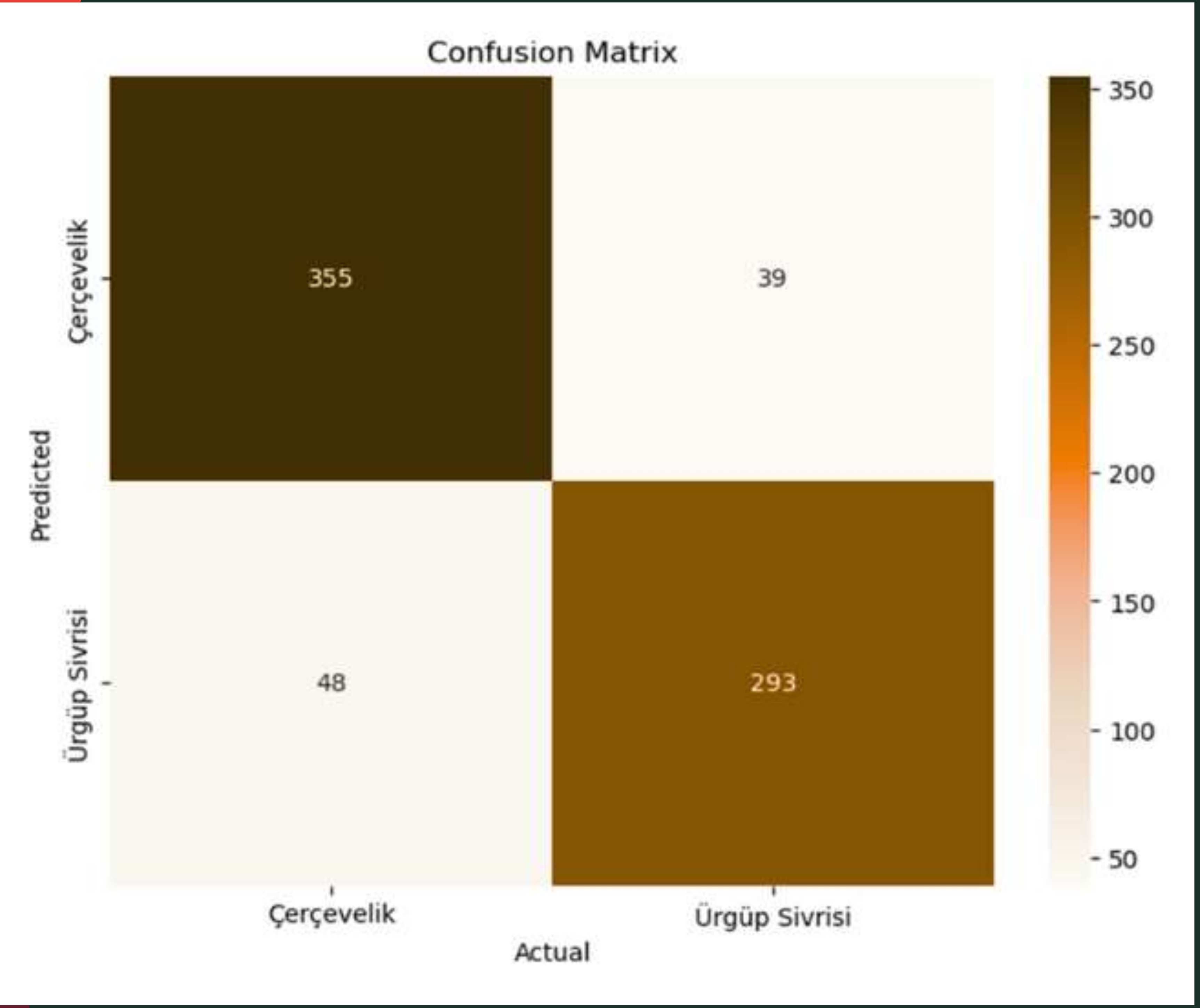
88.16%

CONFIUSON MATRXI

Gradient Boosting (GBN)

Accuracy
88.16 %

F1 score
88.16 %



CONFIUSON MATRXI

NERUL NETWORK (NN)

Accuracy

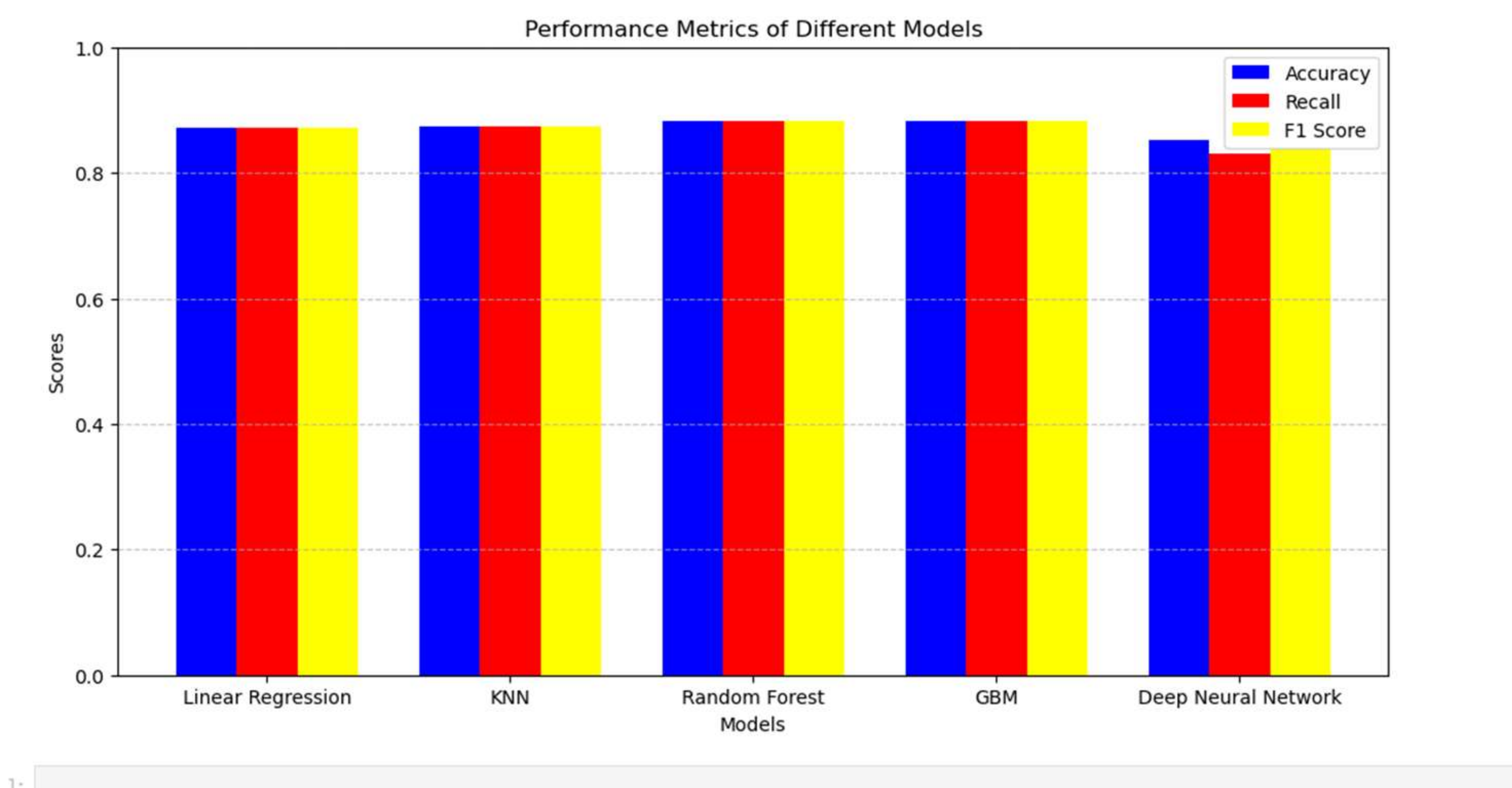
89.12 %

F1 score

87.95 %

```
23/23 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step
Test accuracy: 0.8912, Recall: 0.8563, F1 Score: 0.8795
Training with optimizer=adam, units_1=32, units_2=64, epochs=30, batch_size=16
```

Analysis



Performance Metrics of Different Models
1.0
0.8
0.6
0.4
0.2
0.0
Scores
Accuracy
Recall
F1 Score
Linear Regression
KNN
Random Forest
GBM
Deep Neural Network
Models

SIMILAR RESULT OF MODELS

- No Outliers or Missing Values

- Balanced Features & Labels: The dataset had well-distributed classes and features, reducing model performance differences.

- Feature Selection: Removing redundant features reduced unnecessary complexity, allowing simpler models to perform well.



Business Value




Automated Sorting & Quality Control

Reduced Labor Costs

2,000 metric tons/year of pumpkin seeds

Manual

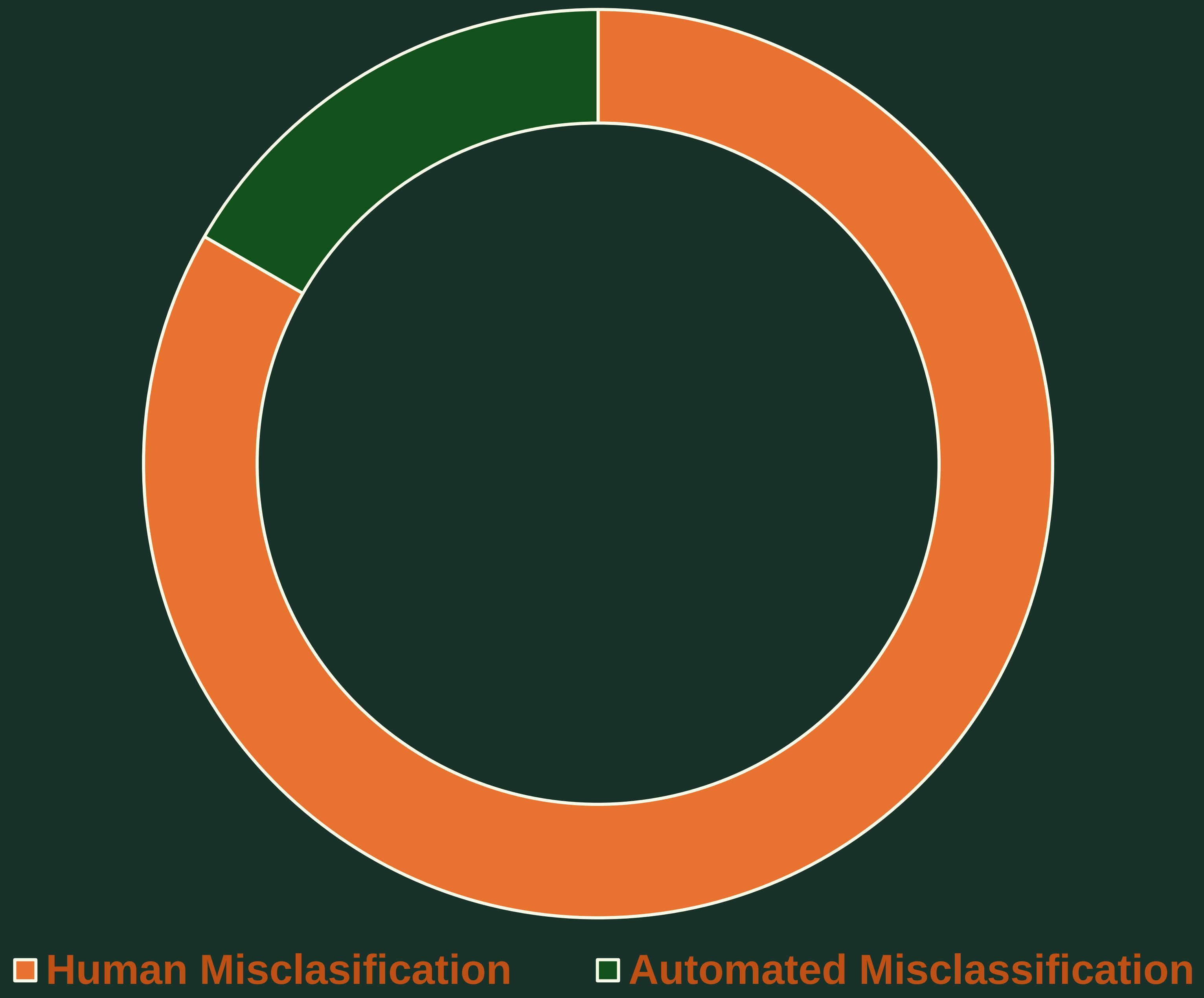
15*30,000= $450,000/year

Automated

3*30,000= $90,000/year

Labor Savings: $360,000/year

Lower Error Rates



Premium Branding & Seed Purity Guarantees

Price premium

Distributor sells 500 tons of “Type A” seeds annually at $2,000 per ton (total $1.0 million)

By Premium quality versus, standard competitor

Price rises to $2,100 per ton

Additional Revenue: $100 × 500 tons = $50,000/year.

Reduced Mislabeling Risk and Fewer Refunds

By blending **premium pricing** with **lower refunds**

Agronomic Consulting and Data Analytics

Data-Driven Yield Improvement

The system can flag anomalies in seed shape or size distribution, indicating a potential issue on the farm (irrigation, fertilization, disease).

Maturity indices

Data

Track seed quality

Shape uniformity

If a farm's total harvest is worth $5 million, and data analytics helps avoid or mitigate a **5%** crop loss, that saves $250,000 in a single season.

Tiered Product Lines

- "Premium"
- "standard"
- "Bulk"

OTHER USECASES

- Development of smart machines for factories
- Possibility to identify other seed varieties
- Quality control line of factory

Conclusion

This project successfully classifies pumpkin seeds by analyzing 2,500 samples and implementing machine learning algorithms to develop a model that can be used to improve:

This project successfully classifies pumpkin seeds by analyzing 2,500 samples and implementing machine learning algorithms to develop a model that can be used to improve:

- High Accuracy Classification
- Improved Efficiency
- Business Benefits
- Broader Agricultural Applications
- Optimized Agricultural Processes

References

📌DATASET:
https://www.muratkoklu.com/datasets/

📌https://doi.org/10.1007/s10722-021-01226-0
📌https://link.springer.com/article/10.1007/s10722-021-01226-0

📌Paper:
https://link.springer.com/content/pdf/10.1007/s10722-021-01226-0.pdf

Thank You!